

Solutions to Quiz 3

Question 1

Given that 5 is an element of multiplicative order 6 in $\mathbb{Z}/\langle 7 \rangle$, Find an integer a that multiplicative order 6 in $\mathbb{Z}/\langle 7^3 \rangle$.

Solution 1:

We are *given* that $5^7 = 5 + 7k_1$ for some integer k_1 . As shown in class, this means $5^{7^{r+1}} = 5^{7^r} + 7^{r+1}k_r$ for each $r \geq 0$, for some integers k_r .

We have also seen that this means that $(5^{7^n})_n$ converges in $\mathbb{Z}_7$ to a 6-th root of unity w .

$$w = 5 + \sum_{r=1}^{\infty} (5^{7^{r+1}} - 5^{7^r})$$

Given that 5 has multiplicative order 6 modulo 7 and $w - 5$ is in $7\mathbb{Z}_7$, we see that w has order *precisely* 6.

Thus, we want an integer a such that $w - a$ lies in $7^3\mathbb{Z}_7$. By the above series, we see that we only need to sum the first 2 terms to get

$$a = 5 + \sum_{r=1}^2 (5^{7^{r+1}} - 5^{7^r}) = 5^{7^3}$$

Question 2

Find an integer $a = p + 5q + 25r$ with $p, q, r \in \{0, \dots, 4\}$ such that $7a - 1$ is divisible by 5^3 .

Solution 2:

We note that $7m - 1 = 5n$ has the solution $(m, n) = (3, 4)$.

It follows that $(7m - 1)^3$ is divisible by 5^3 where $m = 3$.

By the Binomial Theorem we see that

$$(7m - 1)^3 = 7^3m^3 - 3 \cdot 7^2m^2 + 3 \cdot 7m - 1 = 7a - 1$$

where $a = 7^2m^3 - 3 \cdot 7m^2 + 3m$. Since all the terms are divisible by $m^2 = 3m = 9$, we get

$$a/9 = 49 \cdot 3 - 3 \cdot 7 + 1 \equiv 22 - 21 + 1 \pmod{125}$$

This gives $a = 18 = 3 + 5 \cdot 3 + 5^2 \cdot 0$.

So $(p, q, r) = (3, 3, 0)$.

Alternatively, one could note that $125 + 1 = 126$ is divisible by 7. In fact $126 = 7 \cdot 18$ which gives $a = 18$ as before.